Helena Fu

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Education

Tufts University - School of Engineering Medford, MA
B.S. in Computer Science, B.S. in Mathematics (GPA: 3.93 / 4.00) *May 2027*

Honors: 4x Tufts Dean's List, Al Gardner Memorial Endowment Scholarship (Society of Women Engineers).

Relevant Coursework: Machine Learning Foundations, Linear Algebra, Differential Equations, Data Structures, Mathematical Aspects of Data Analysis, Algorithms, Mathematical Modeling, Statistical Pattern Recognition.

Technical Skills

Machine Learning/AI: scikit-learn, PyTorch, TensorFlow, Keras, BERT, LSTM, RL, HMM, OpenCV, Claude, Gemini
Languages: Java, Python, C/C++/C#, SQL, JavaScript, HTML/CSS, R, Assembly, ROS, Bash Script, MatLab, OCaml, PHP
Frameworks: React, Node.js, Flask, WordPress, Next.JS
Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, AWS Bedrock/Lambda
Libraries: pandas, NumPy, Matplotlib, HuggingFace

Experience

Human-Robot Interaction Lab at Tufts University Feb. 2024 – May. 2025
Research Assistant *Medford, MA*

- Built real-time pupillometry preprocessing pipeline in Python; boosted data quality for cognitive state ML models by **30%**.
- Integrated **NLP system** with DIARC robot architecture to enable unstructured commands from **50+** study participants.
- Implemented natural language cache to store and retrieve prior LLM responses, reducing latency by **46%**.

MIT Schwarzman College of Computing Aug. 2024 – Dec. 2024
Machine Learning Intern *Cambridge, MA*

- Preprocessed **11GB** of unstructured sensor and biochemical data from **3** marine data sources.
- Developed **multi-modal ML pipeline** for hydrographic prediction; achieved **97%** regression accuracy.
- Engineered features and applied ensemble models to predict harmful algal bloom risk with **76%** accuracy.

Lawrence Berkeley National Laboratory June 2023 – July 2023
Research Intern *Berkeley, CA*

- Trained and benchmarked **100+ CNN architectures** for heterogeneous hardware image classification optimization.
- Achieved **83%** accuracy on CIFAR-10 dataset using hyperparameter tuning and dropout regularization.

Projects

ML Quantitative Trading | *The Herd Project Selection* Jan. 2025 – Present

- Designed automated trading bot using **RL, LSTM, HMM,** and **BERT**; improved ROI by **53%** over moving average baseline.
- Built real-time inference system with live price feeds, retraining models with updated market trends.

DSCVR.AI | *Cornell NYC AI* ***Hackathon Winner*** Feb. 2025 – Mar. 2025

- Posttrained foundation model using domain-specific prompts via **Claude API**, **Docker**, and **AWS Bedrock**.
- Built AI-assisted writing platform with LLM-driven autocomplete; engineered seamless UX with **Next.js** and **Tailwind**.
- Earned **Best UX**, and **Most App Store Ready** awards for integrated ML experience.

AJL Kaggle Competition | ***6th*** *Place out of* ***350+*** Jan. 2025 – Present

- Evaluated Vision Transformer variants (DeiT, Swin, Google ViT, ConvNeXt V2) via **TorchVision** on Fitzpatrick17k dataset.
- Achieved **71%** on image classification task with custom **PyTorch** training pipeline using **CrossEntropyLoss** and **Adam**.
- Awarded top prize for novel image preprocessing algorithms improving model robustness.